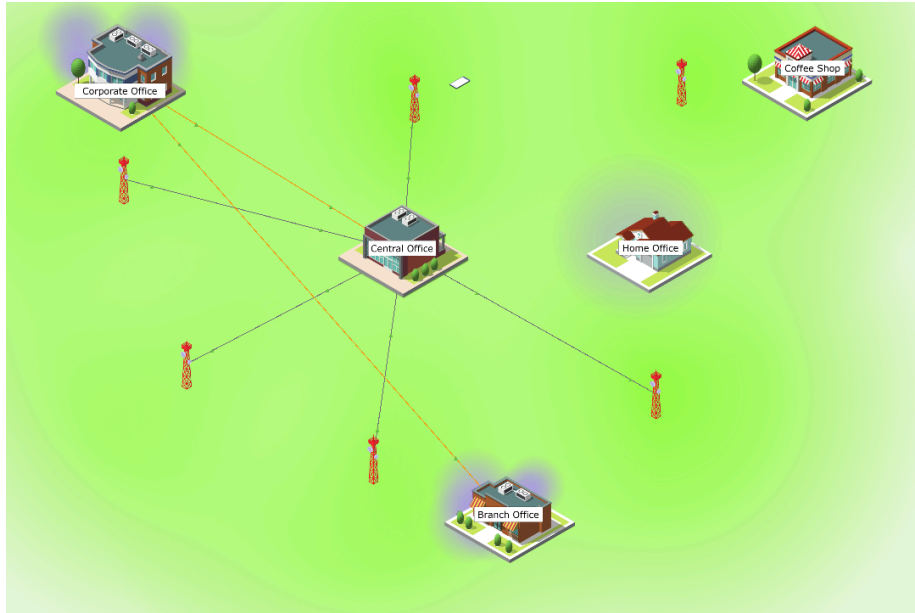


Lab7: 13.5.2 Packet Tracer - Wireless Technology Exploration - Physical Mode

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Corporate Office				
Corporate Router	G1/0	192.168.1.1	255.255.255.0	N/A
	G2/0	192.168.4.1	255.255.255.252	
	G3/0	192.168.3.1	255.255.255.252	
Corporate Server	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
Corporate WLC	G2	192.168.1.50	255.255.255.0	192.168.1.1
LAP-C1 to C6	G0	DHCP	N/A	N/A
Laptop	Fa0	DHCP	N/A	N/A
Branch Office				
Branch Router	G0/0	192.168.2.1	255.255.255.0	N/A
	G2/0	192.168.4.2	255.255.255.252	
Branch Server	Fa0	192.168.2.10	255.255.255.0	192.168.2.1
Branch WLC	G2	192.168.2.50	255.255.255.0	192.168.2.1
LAP-B1 to B6	G0	DHCP	N/A	192.168.2.1
Central Office (CO)				
CO Server	N/A	192.168.5.2	255.255.255.252	192.168.5.1
Cellular	N/A	172.16.1.1	255.255.255.0	N/A
CO Router	G0/0	192.168.5.1	255.255.255.252	N/A
	G1/0	192.168.6.1	255.255.255.252	
	G3/0	192.168.3.2	255.255.255.252	
Cell Towers 0-5	Coaxial 0-2, 4-5	N/A	N/A	N/A



Tasks

Objectives

Part 1: Explore the Wireless Network

Part 2: Add Wi-Fi Connectivity to a Boardroom

Part 3: Add Wireless Connectivity to a Coffee Shop in a Cellular Dead Zone

Part 4: Add Wireless Connectivity to a Home Office

Background / Scenario

XYZ Corporation is expanding their network capabilities to allow enhanced connectivity at their local offices, as well as connectivity for those wishing to work remotely. In this Packet Tracer Physical Mode (PTPM) activity, you have been asked to assist with this plan by reviewing the current network capabilities and adding wireless functionality as required.

Note: This activity is not graded. Your connectivity tests can be used to validate your physical connections and configurations.

Instructions

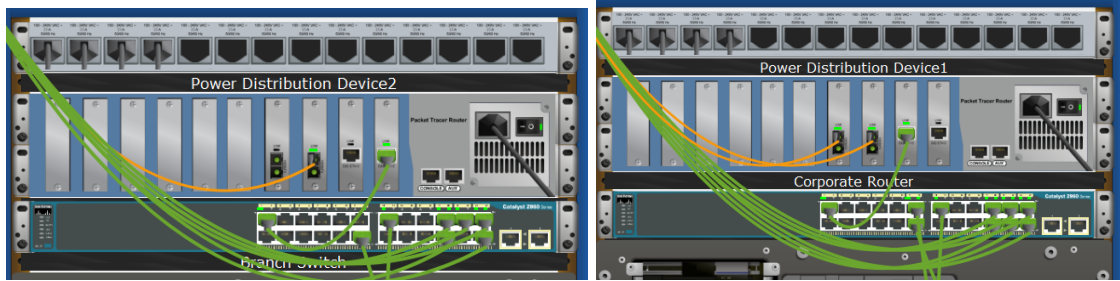
Part 1: Explore the Wireless Network

In Part 1, you will explore the wireless network and verify connectivity.

Note: Wireless and cellular signal representations are **on** in this activity. However, you can turn them off by clicking **View Wireless Signals** (Ctrl+Shift+W) on the top blue toolbar.

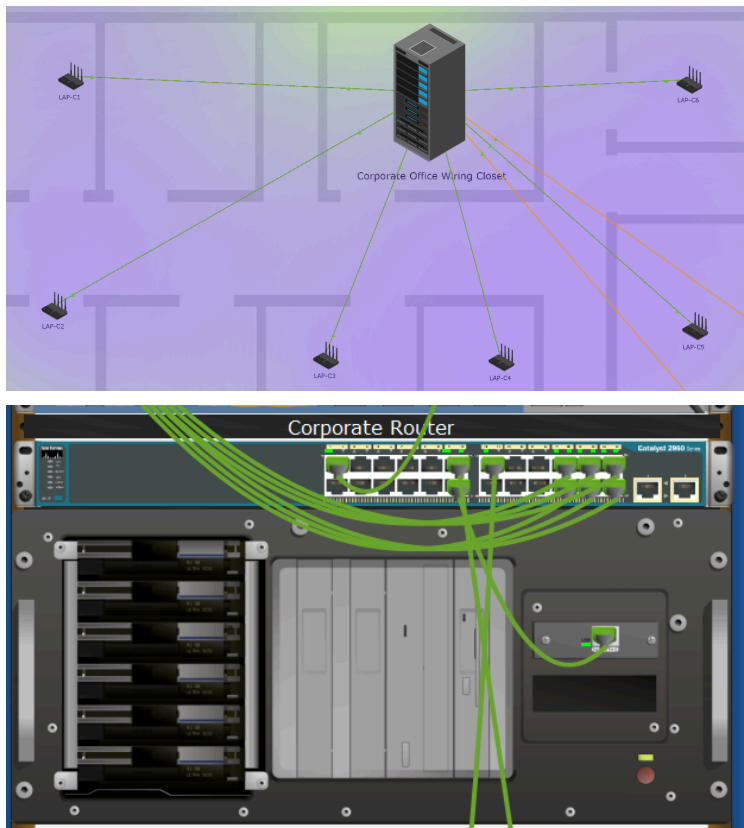
Step 1: Explore the topology.

- In **Physical** mode, you will notice that **Home City** contains five different locations: a **Corporate Office**, a **Branch Office**, a **Central Office (CO)**, a **Home Office**, and a **Coffee Shop**. How are the four offices connected and what type of cables connect them?



The Corporate Office and Branch Office are connected to the Central Office through fiber-optic links. The Home Office connects through a cable modem using a coaxial cable, while wireless devices inside the Home Office access the network through a wireless router.

- b. Navigate inside the **Corporate Office**. Notice that six Lightweight Wireless Access Points (LWAPs) are connected to the wiring closet.



How are the LWAPs connected to the network?

The LWAPs are connected to the network through the bundle of green Ethernet cables running off-screen to the top-left.

Tracing those cables back to the rack, they are all plugged directly into the ports of the Switch (located right below the Corporate Router).

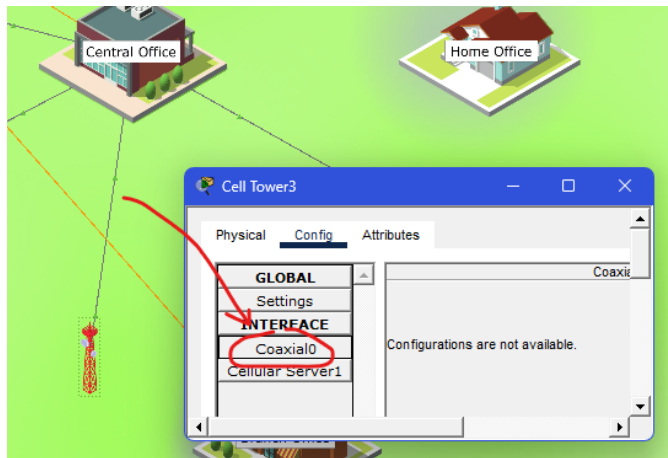
From there, the switch manages their local network connectivity and links them back up to the rest of the corporate network infrastructure

c. Navigate into the CO wiring closet.

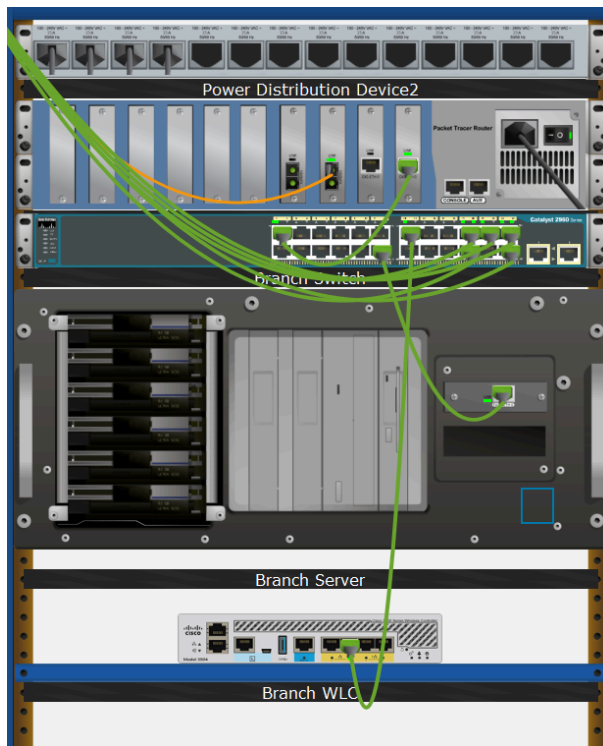
How are the cell towers connected to the **Central Office Server**?

The PT-CELL-NM-1CX card features a single coaxial connector, which is used for a cable modem service connection

This show that it transmits and receives signals via coaxial cable, as seen in the city plan.

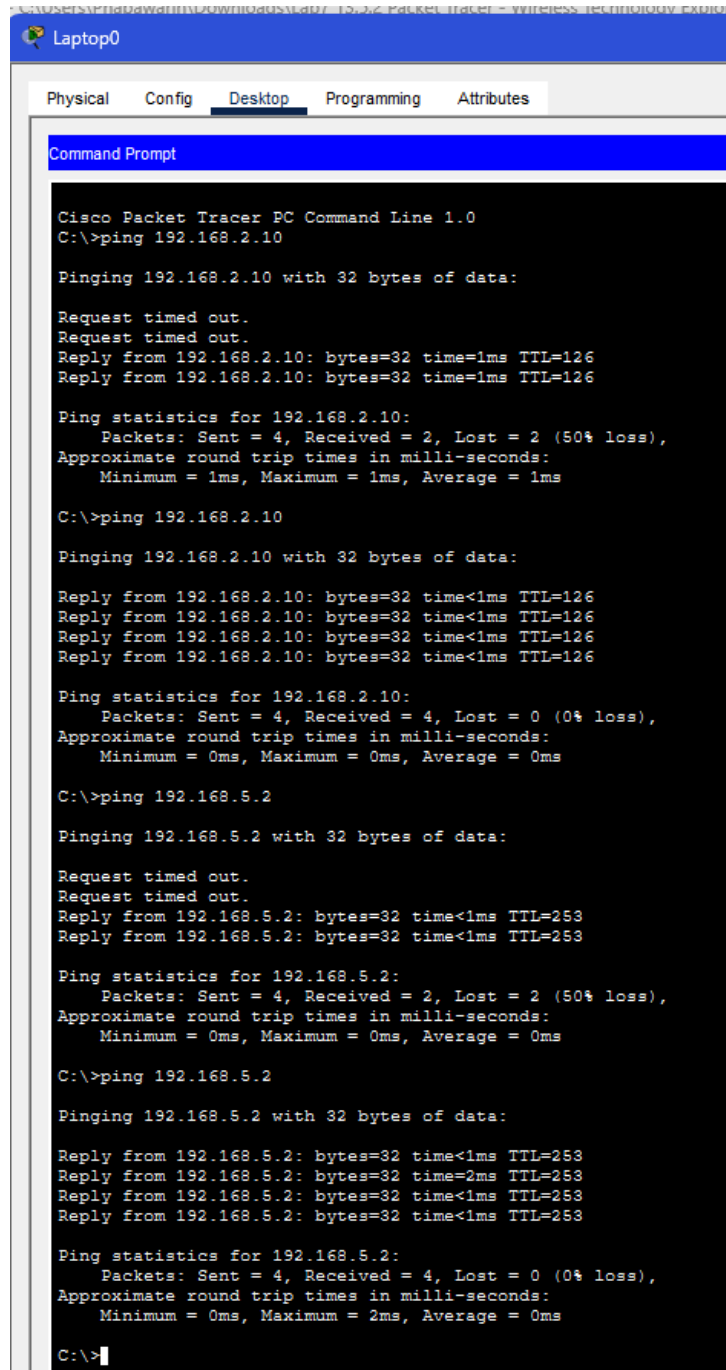


d. Navigate to the **Branch Office**. Notice that five LWAPs are connected via copper cable to the **Branch Switch** in the wiring closet. The switch is then connected to the **Branch WLC** (wireless LAN controller).



Step 2: Verify connectivity.

- a. To verify connectivity, navigate to **Corporate Office > Corporate Office Wiring Closet**.
- b. Click the laptop, and then **Desktop** tab > **Command Prompt**.
 - 1) Ping 192.168.2.10 (the **Branch Office Server**).
 - 2) Ping 192.168.5.2 (the **Central Office** backbone connection).



```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.5.2: bytes=32 time<1ms TTL=253
Reply from 192.168.5.2: bytes=32 time<1ms TTL=253

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time<1ms TTL=253
Reply from 192.168.5.2: bytes=32 time=2ms TTL=253
Reply from 192.168.5.2: bytes=32 time<1ms TTL=253
Reply from 192.168.5.2: bytes=32 time<1ms TTL=253

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

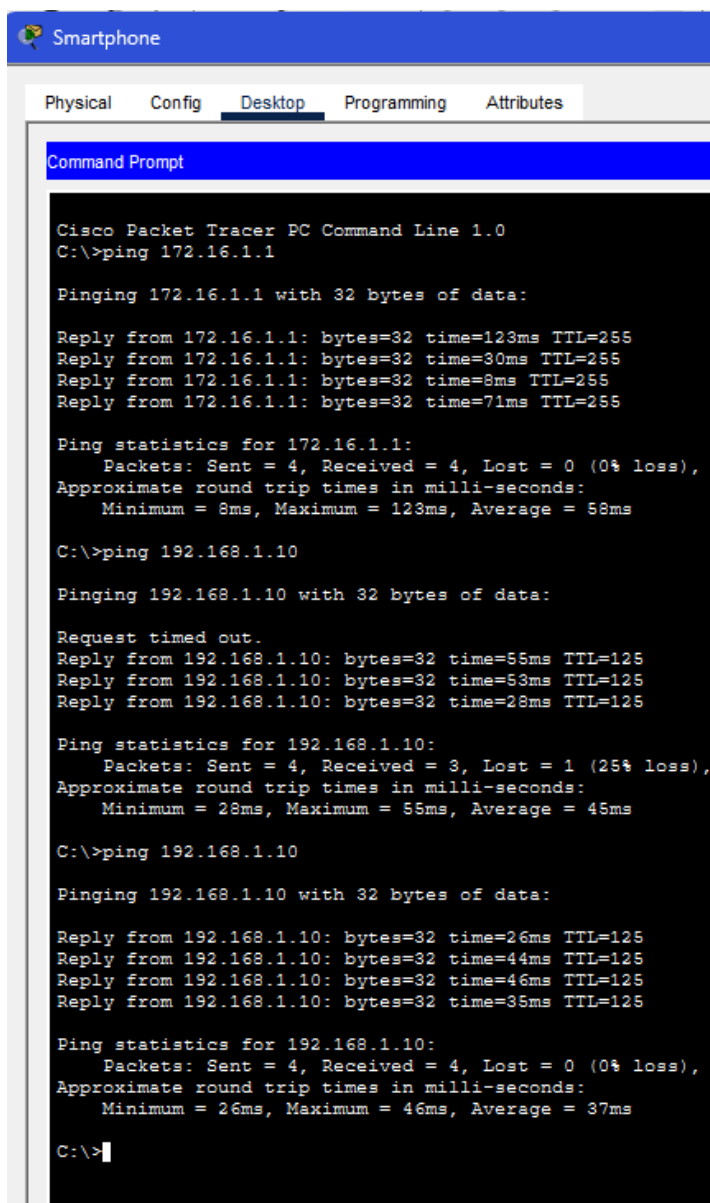
These two are 50% loss in the first, then the next time is 0% loss. Maybe it need some times.

Note: Packet Tracer can take some time to converge. You may receive **Request time out** messages. However, both of the pings should eventually be successful. At the bottom of the Packet Tracer interface, click **Fast Forward Time** several times to speed up convergence.

Yup! I tried that by ping it two times.

- c. Navigate to the **Home City**. Click the **Smartphone** next to the cell tower above the **Central Office** building.
- d. Click **Desktop** tab, and then **Command Prompt**.
 - 1) Ping 172.16.1.1 (the cellular connection at the **Central Office**).
 - 2) Ping 192.168.1.10 (the **Corporate Office Server**).

You may receive one or two **Request time out** messages. However, both of the pings should be successful.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time=123ms TTL=255
Reply from 172.16.1.1: bytes=32 time=30ms TTL=255
Reply from 172.16.1.1: bytes=32 time=8ms TTL=255
Reply from 172.16.1.1: bytes=32 time=71ms TTL=255

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 123ms, Average = 58ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.10: bytes=32 time=55ms TTL=125
Reply from 192.168.1.10: bytes=32 time=53ms TTL=125
Reply from 192.168.1.10: bytes=32 time=28ms TTL=125

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 55ms, Average = 45ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=26ms TTL=125
Reply from 192.168.1.10: bytes=32 time=44ms TTL=125
Reply from 192.168.1.10: bytes=32 time=46ms TTL=125
Reply from 192.168.1.10: bytes=32 time=35ms TTL=125

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 46ms, Average = 37ms

C:\>
```

What are the different physical connections used between **Smartphone1** and the **Corporate Office Server**?

The three different types of physical connections is

Wireless: Starting from Smartphone1 Then the device connects wirelessly via a cell tower or Access Point to the network.

Copper Straight-Through: Once the signal reaches the network device, the data travels through the green Ethernet cable used to connect the Access Point to the Switch, and from the Switch to the Corporate Office Server.

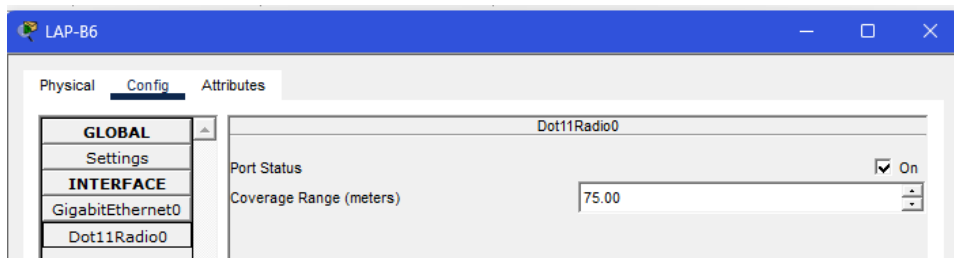
Fiber Optic: This orange cable connects the ports of the Router (Corporate Router) to transmit high-speed data across branches or the main connection point in the city (Central Office).

Part 2: Add Wi-Fi Connectivity to a Boardroom

A new boardroom is being created within the **Branch Office**. Currently the boardroom is in a Wi-Fi dead zone. In Part 2, your task is to provide connectivity for devices within that room.

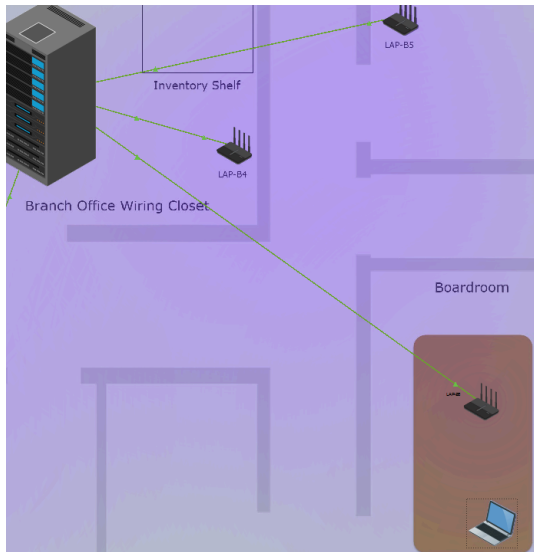
Step 1: Install a new LAP-PT device to provide coverage for the new boardroom.

- Navigate to the **Branch Office**. The laptop inside the new boardroom has no access to a Wi-Fi signal.
- Click and drag the access point on the **Inventory Shelf** to the **Boardroom**.
- Click the new access point to open it. From the **Modules** menu, click and drag an **ACCESS_POINT_POWER_ADAPTER** and connect it to the power port next to the **Reset** button.
- Click the **Config** tab. Under **GLOBAL Settings** name the device **LAP-B6**.
- Under **INTERFACE Dot11Radio0** set the **Coverage Range** to **75.00** meters. Packet Tracer does not grade this setting.



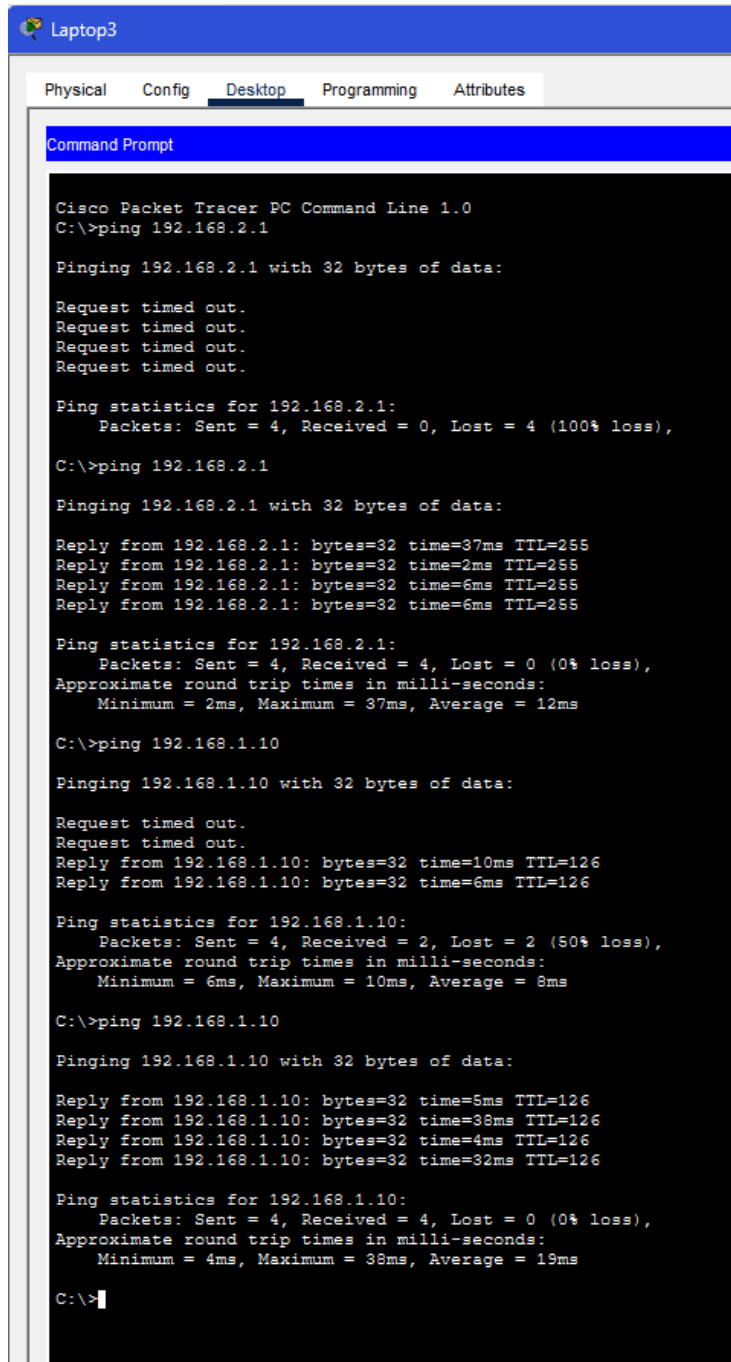
- Close the window for **LAP-B6**. If you have **View Wireless Signals** on, you will notice that there is now coverage in the boardroom.
- From the **Bottom Toolbar**, click **Connections > Copper Straight-Through** cable.
- Connect one end of the cable to the **GigabitEthernet0** interface on the new access point. Click the rack of equipment and connect the other end of the cable to the **Rack > Branch Switch > Fa0/22 interface**.

For anybody who needs to configure the LAP. Please do not press the reset button like me!! 😬
It's make the laptop couldn't ping Corporate Server and Default Gateway.



Step 2: Verify Connectivity.

- a. In the boardroom, click the laptop, and then **Desktop** tab > **IP Configuration**. The laptop should now have full IPv4 configuration under **IP Configuration**. However, the DHCP processes may take a few minutes to complete. If necessary, toggle between DHCP and Static to resend a DHCP request. You may also need to click **Fast Forward Time** several times to speed up convergence.
- b. When the IP addressing is provided, you can verify connectivity. Close the **IP Configuration** window, and then click **Command Prompt**.
 - 1) Ping 192.168.2.1 (the **default gateway**).
 - 2) Ping 192.168.1.10 (the **Corporate Server**).



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=37ms TTL=255
Reply from 192.168.2.1: bytes=32 time=2ms TTL=255
Reply from 192.168.2.1: bytes=32 time=6ms TTL=255
Reply from 192.168.2.1: bytes=32 time=6ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 37ms, Average = 12ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.10: bytes=32 time=10ms TTL=126
Reply from 192.168.1.10: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 10ms, Average = 8ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=5ms TTL=126
Reply from 192.168.1.10: bytes=32 time=38ms TTL=126
Reply from 192.168.1.10: bytes=32 time=4ms TTL=126
Reply from 192.168.1.10: bytes=32 time=32ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 38ms, Average = 19ms

C:\>
```

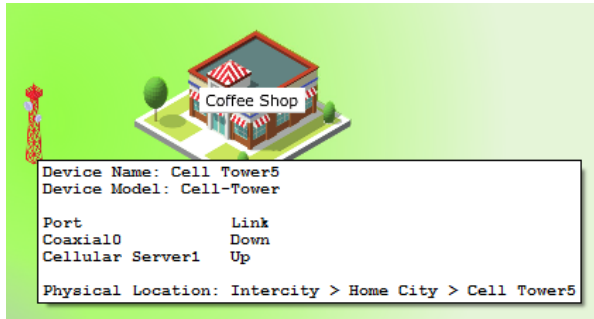
struggle to succeed' 😞...

Part 3: Add Wireless Connectivity to a Coffee Shop in a Cellular Dead Zone

A new **Coffee Shop** has opened in the **Home City**, but there is currently no cellular service in this area. Your task is to provide cellular service for customers and employees of the coffee shop.

Step 1: Connect a new cell tower.

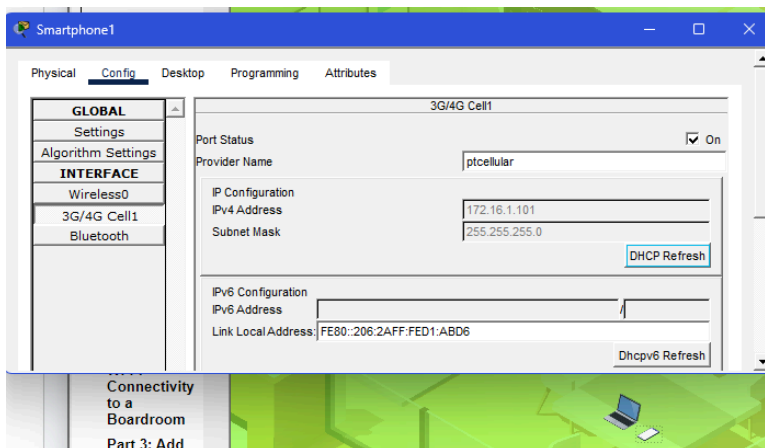
- Navigate to the **Home City**.
- Notice that the **Cell Tower** next to the **Coffee Shop** is not connected to the **Central Office**.



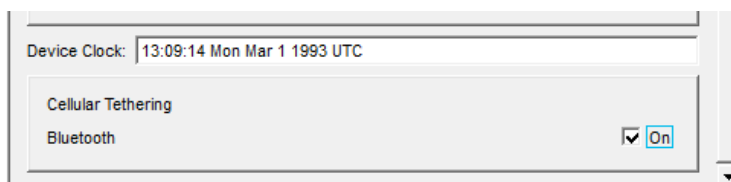
- c. From the Bottom Toolbar, click Connections > Coaxial cable.
- d. Connect one end to the Coaxial0 interface on the unconnected Cell Tower. Then click Central Office > Central Office Wiring Closet > Table > Central Office Server > Coaxial0/3 interface.

Step 2: Tether a laptop to the smartphone.

- a. A remote worker wants to work at the coffee shop. Navigate to the **Coffee Shop** and locate the smartphone and laptop on the table.
- b. Click **Smartphone** > **Config** tab > **3G/4G Cell1** to verify that the smartphone has received an IP address. It may take a few seconds to get addressing information. Click **DHCP Refresh**, if necessary.



- c. Click **Settings** and verify the smartphone has received a default gateway and DNS server address.
- d. Under **Cellular Tethering**, enable Bluetooth.



- e. Under **INTERFACE**, click **Bluetooth**, and set **Port Status** to **On**. Verify that **Discoverable** is enabled.

Bluetooth

Port Status ☒ On

MAC Address 0050.0F85.C73B

Coverage Range (meters) 10.00

Discoverable ☒ On

- f. In the Coffee Shop, click Laptop > Desktop tab > Bluetooth, and set Port Status to On.
- g. Select **Discover** to display **Smartphone1** under discoverable devices.

Laptop1

Physical Config **Desktop** Programming Attributes

Bluetooth Configuration X

Port Status ☒ On

MAC Address 000A.41D7.D1EC

Coverage Range (meters) 10.00

Discoverable ☒ On

Devices

Name	MAC Address	Status
Smartphone1	0050.0F85.C73B	Unpaired

Discover Pair Unpair Tether Untether

- h. Select the **Smartphone1**, click **Pair**, and then answer **Yes** to the **Bluetooth Pairing** popup message.
- i. Click the laptop again, reselect the **Smartphone1**, and then click **Tether**. It may be necessary to move the smartphone and laptop close together for Bluetooth pairing to work.

Laptop1

Physical Config **Desktop** Programming Attributes

Bluetooth Configuration X

Port Status ☒ On

MAC Address 000A.41D7.D1EC

Coverage Range (meters) 10.00

Discoverable ☒ On

Devices

Name	MAC Address	Status
Smartphone1	0050.0F85.C73B	Paired, Connected, Tether Access ...

Discover Pair Unpair **Tether** Untether

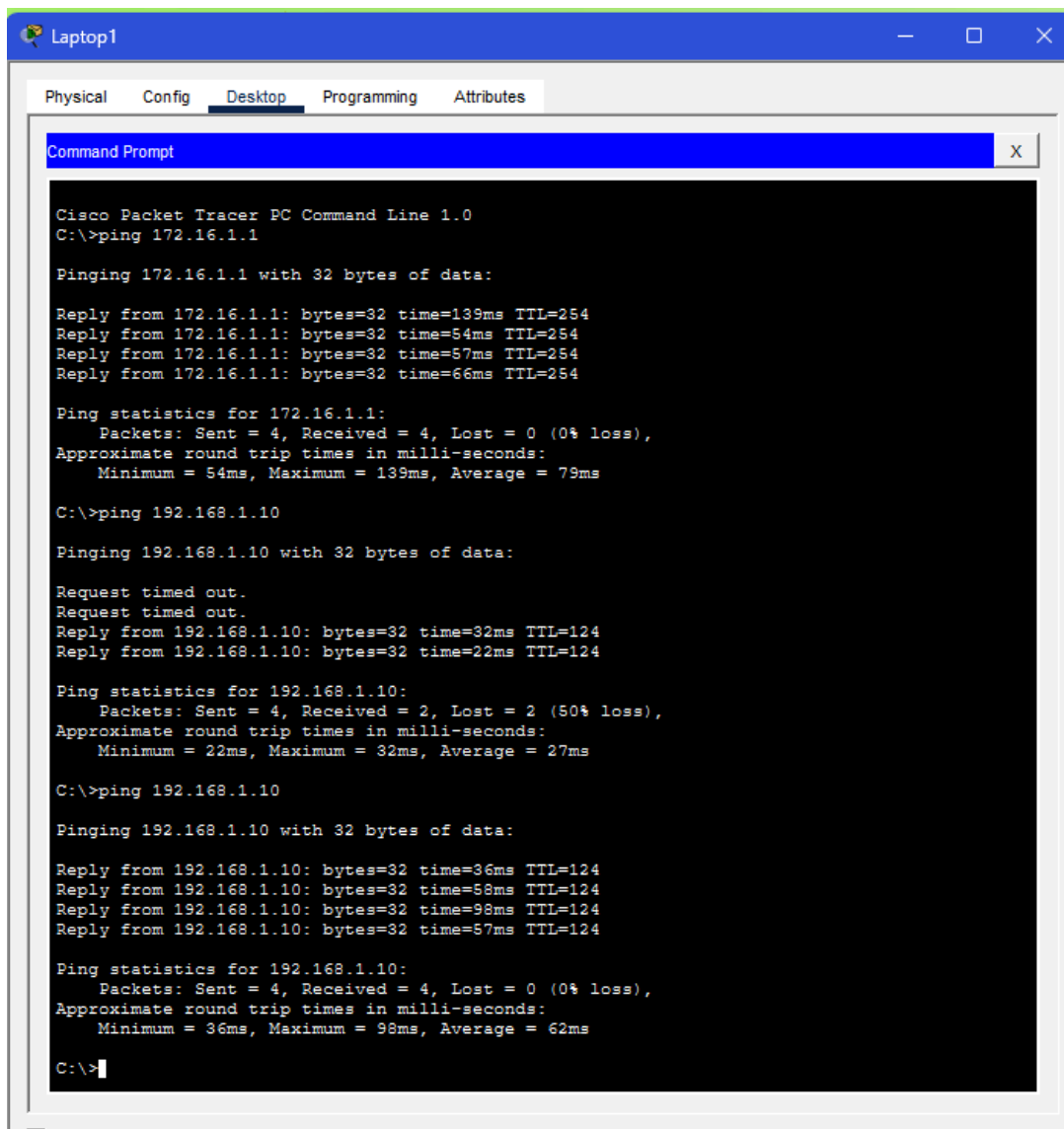
- j. After a few seconds, you should see valid addressing information in the **IP Configuration** section. If not, repeat the previous steps.

IP Configuration

IP Address 172.20.10.2

Subnet Mask 255.255.255.240

k. To verify connectivity, close the **Bluetooth Configuration** window, and then click **Command Prompt**. Ping the **cellular gateway** (172.16.1.1) and the **Corporate Office Server** (192.168.1.10). If the first ping to the **Corporate Office Server** is not successful, try another ping.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time=139ms TTL=254
Reply from 172.16.1.1: bytes=32 time=54ms TTL=254
Reply from 172.16.1.1: bytes=32 time=57ms TTL=254
Reply from 172.16.1.1: bytes=32 time=66ms TTL=254

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 54ms, Maximum = 139ms, Average = 79ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.10: bytes=32 time=32ms TTL=124
Reply from 192.168.1.10: bytes=32 time=22ms TTL=124

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 32ms, Average = 27ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=36ms TTL=124
Reply from 192.168.1.10: bytes=32 time=58ms TTL=124
Reply from 192.168.1.10: bytes=32 time=98ms TTL=124
Reply from 192.168.1.10: bytes=32 time=57ms TTL=124

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 98ms, Average = 62ms

C:\>
```

The Corporate Office Server seems lazy again. It needs a second try to ping it.

Part 4: Add Wireless Connectivity to a Home Office

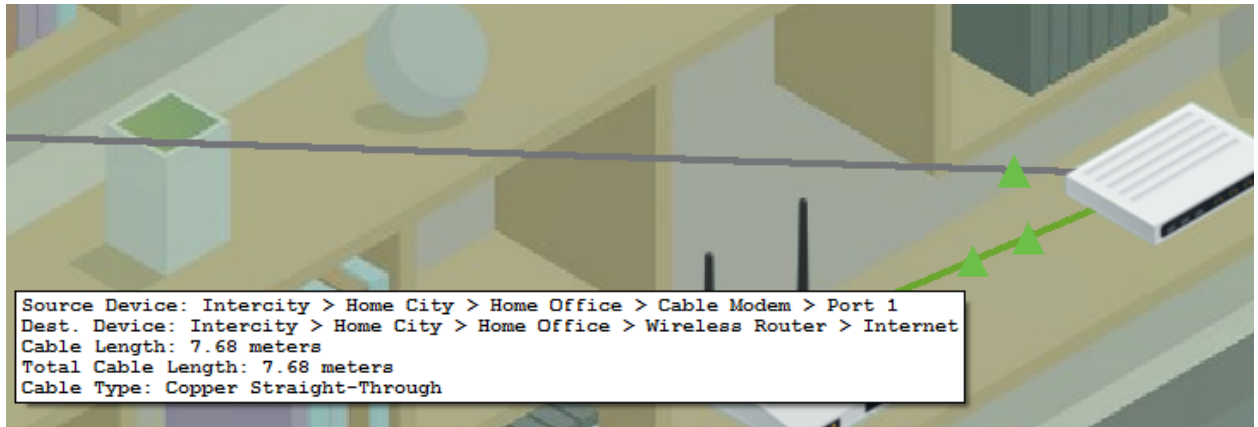
A remote worker for XYZ Corporation has just moved and the new house does not have a network setup yet. Your task is to set up a network to provide wireless access throughout the house and connectivity to the Corporate Office.

Step 1: Select and cable the devices.

a. Navigate to the **Home City**, and then **Home Office**.

On the shelf behind the desk chair, there is a wireless router with external antennae. There is also a cable modem directly to the right of the wireless router. On the table in front of the couch, there is a laptop.

- b. Click **Connections** > **Copper Straight-Through**.
- c. Connect one end of the cable to **Port 1** of **Cable Modem**. Connect the other end to the **Internet** port on **Wireless Router**.



- d. Navigate to the **Home City** view.
- e. From the **Bottom Toolbar**, click **Connections** > **Coaxial** cable.
- f. Click **Home Office** > **Cable Modem0** > **Port 0**, and then click the **Central Office** > **Central Office Wiring Closet** > **Rack** > **CMTS** > **Coaxial7** port.



Step 2: Configure the wireless router.

- a. Navigate to the **Home Office**, and click **Wireless Router** > **GUI** tab.
- b. The **Setup** tab is already selected. For **Internet Connection Type**, verify **Automatic Configuration - DHCP** is selected.
- c. Under **Network Setup**, verify that the following information is configured:

IP Address: **192.168.0.1**

Subnet Mask: **255.255.255.0**

DHCP: **Enabled**

Starting IP Address: **192.168.0.100**

Maximum number of users: **50**

Wireless Tri-Band Home Router

Firmware Version: v0.9.7

Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Administration	Status
Basic Setup	DHCP	MAC Address Filter	Guest Network	Wireless MAC Filter	Advanced Routing	HomeRouter-PT-AC

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers):

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: . . .

Subnet Mask:

DHCP Server Settings

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.0.

Maximum number of Users:

IP Address Range: 192.168.0. 100 - 149

Client Lease Time: minutes (0 means one day)

Static DNS 1: . . .

Static DNS 2: . . .

Help...

- d. Scroll back to the top and click the **Status** tab. Under **Internet Connection**, the wireless router should have DHCP addressing from the **Central Office**. If not, click **IP Address Renew** to resend DHCP messages.
- e. Click the **Wireless** tab.
- f. Configure the 2.4 GHz network with **homesweethome** as the **Network Name (SSID)**. Scroll to the bottom and click **Save Settings**.
- g. Scroll back to the top and select the **Wireless Security** subtab.
- h. For the 2.4 GHz **Security Mode**, select **WPA2-Personal**, and then configure **mySecureNet** as the **Passphrase**. Scroll to the bottom and click **Save Settings**.

Wireless Tri-Band Home Router

Firmware Version: v0.9.7

Wireless	Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Administration	Status
Basic Wireless Settings	Wireless Security	Guest Network	Wireless MAC Filter	Advanced Wireless Settings	HomeRouter-PT-AC		

Wireless Security

2.4 GHz

Security Mode:

Encryption:

Passphrase:

Key Renewal: seconds

5 GHz - 1

Security Mode:

5 GHz - 2

Security Mode:

Help...

Step 3: Verify connectivity.

- On the table in front of the couch, click the laptop, and then the **Config** tab. Next, select **Wireless0** under **INTERFACE**.
- Enter the SSID **homesweethome**.
- Select **WPA2-PSK** for the **Authentication** method, and then configure **mySecureNet** as the **PSK Pass Phrase**.
- Under **IP Configuration**, the laptop should receive DHCP addressing. You may need to toggle between DHCP and Static a few times to resend DHCP requests.

Wireless0	
Port Status	☒ On
Bandwidth	200 Mbps
MAC Address	0090.2BE8.CA46
SSID	homesweethome
Authentication	
☐ Disabled	☐ WEP
☐ WPA-PSK	☒ WPA2-PSK
☐ WPA	☐ WPA2
☐ 802.1X	Method: MD5
WEP Key	
PSK Pass Phrase	mySecureNet
User ID	
Password	
User Name	
Password	
Encryption Type	AES
IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	169.254.202.70
Subnet Mask	255.255.0.0

- Click **Desktop** tab > **Command Prompt**. Ping various addresses throughout the network. For example, the following pings should be successful:

ping 192.168.6.1 (**Central Office router G1/0**)

ping 192.168.1.10 (**Corporate Office Server**)

ping 192.168.2.10 (**Branch Office Server**)

Command Prompt

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.6.1

Pinging 192.168.6.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.6.1: bytes=32 time=28ms TTL=254
Reply from 192.168.6.1: bytes=32 time=32ms TTL=254
Reply from 192.168.6.1: bytes=32 time=92ms TTL=254

Ping statistics for 192.168.6.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms, Maximum = 92ms, Average = 50ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=29ms TTL=125
Reply from 192.168.1.10: bytes=32 time=64ms TTL=125
Reply from 192.168.1.10: bytes=32 time=6ms TTL=125
Reply from 192.168.1.10: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 64ms, Average = 27ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=94ms TTL=124
Reply from 192.168.2.10: bytes=32 time=30ms TTL=124
Reply from 192.168.2.10: bytes=32 time=5ms TTL=124

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 94ms, Average = 43ms

C:\>
```

Reflection

1. What overall benefit does wireless technology provide to the end user?

The convenience of wireless LAN connectivity, the ability to move freely to different work locations, and the ease of adding new devices to the network.

2. Which form of wireless networking is the most beneficial?

It depends on your perspective and usage. If speed and stability are your priority indoors/in your home, Wi-Fi is usually the best option. However, if you need nationwide coverage outdoors, cellular/5G would be most beneficial.

3. How could each of the following groups benefit from the various wireless technologies presented in this activity?

= Student: It allows you to access the internet for research, online learning, or submitting assignments from anywhere in the school or at home, without being limited to the computer lab. Plus, you can play online games anywhere.

= Salesperson: Customers can access product information, check stock levels, or send customer purchase orders back to the company instantly via mobile (**cellular**) networks, even while traveling to meet customers in person.

= R&D Engineer: It facilitates the connection and testing of **IoT devices**, the transmission of data from wireless sensors for analysis, and the rapid retrieval of technical data in the laboratory. It also provides easy access to academic references via the website, eliminating the need to go to a computer lab and allowing for research while working.

= Corporate Executive: You can track work, approve important documents, or participate in video conferences from anywhere at any time, whether you're in the boardroom, traveling, or from overseas.